

- 6 (a) electron diffraction/ neutron diffraction** **B1**
- (b) $\frac{1}{2}mv^2 = eV$**
- $$\frac{1}{2} \times 9.11 \times 10^{-31} \times v^2 = 1.60 \times 10^{-19} \times 4800$$
- $$v = 4.1 \times 10^7 \text{ m s}^{-1}$$
- C1**
- $$\lambda = h / mv$$
- $$= 6.63 \times 10^{-34} / (9.11 \times 10^{-31} \times 4.1 \times 10^7)$$
- $$= 1.8 \times 10^{-11} \text{ m}$$
- C1**
A1
- (c) (i) Any two points, 1 mark each** **B2**
- photons are (discrete) packets of energy
 - energy of photons depends on frequency (of EM radiation)
 - electrons can only absorb a single photon (of energy)
 - emission only possible if photon energy is at least the work function
- B1**
- (ii) work function = $hf = 6.63 \times 10^{-34} \times 6.93 \times 10^{14}$** **C1**
- $$= 4.59 \times 10^{-19} \text{ J}$$
- $$= 4.59 \times 10^{-19} / 1.60 \times 10^{-19}$$
- $$= 2.87 \text{ eV}$$
- A1**

